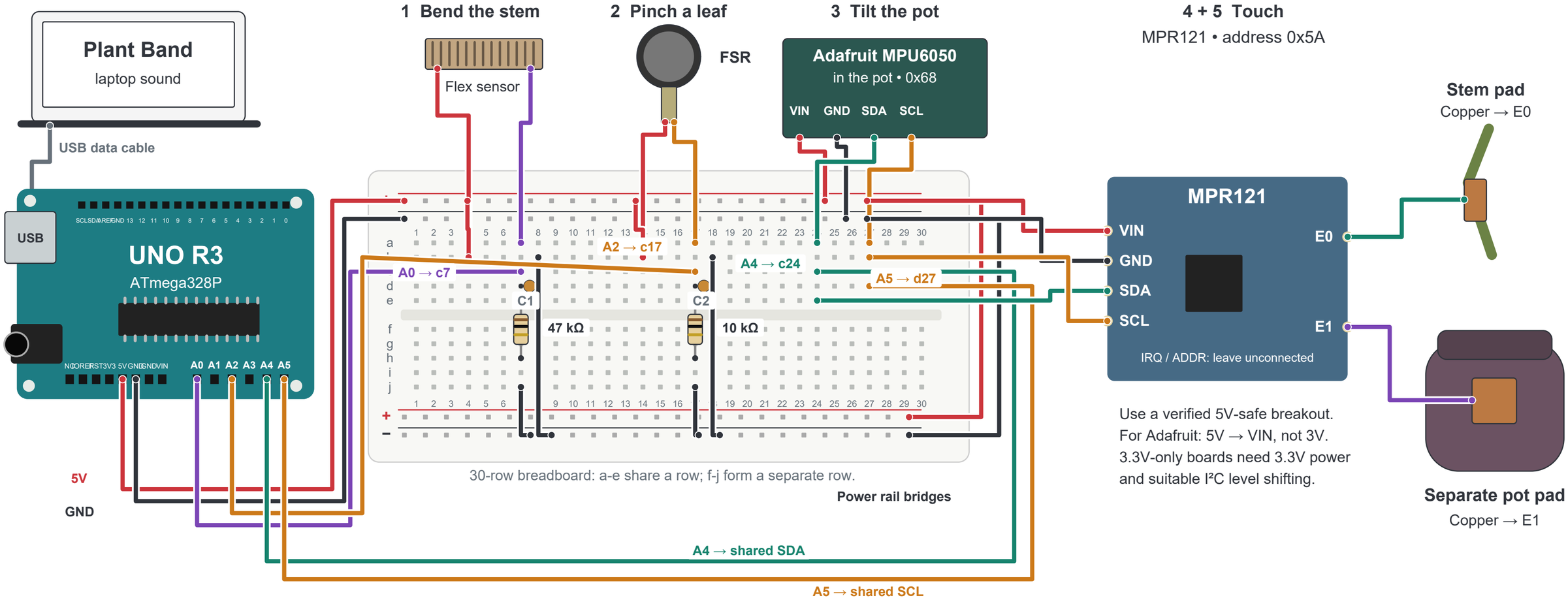
Plant Band

UNO R3 / USB to laptop / five simultaneous inputs

FULL BREADBOARD MAP

Match pin labels on your actual modules.



Build in this order

1. Unplug USB; connect 5V and GND rails.
2. Join the top and bottom power rails.
3. Add the flex and FSR dividers below.
4. Join both sensors to SDA/SCL; add pads.
5. Plug in; Connect Arduino; calibrate.

A1 is unused. No piezo is needed.
Wire ends with dots are the actual connections.
Crossing wires do not join unless a dot is shown.

Exact breadboard holes

Flex: leads a4 / a7. + rail → b4.
A0 → c7. 47 kΩ: e7 → h7. j7 → GND.
C1: d7 → d8; b8 → GND.
FSR: leads a14 / a17. + rail → b14.
A2 → c17. 10 kΩ: e17 → h17. j17 → GND.
C2: d17 → d18; b18 → GND.
SDA: A4 → c24; MPU → a24; MPR → e24.
SCL: A5 → d27; MPU → a27; MPR → b27.

C1 / C2 = 100 nF ceramic (non-polarized).

Before playing

Fix the MPU6050 firmly to a dry part of the pot.
Set upright while still; then set a gentle full tilt.
MPU: VIN → 5V, GND → GND. INT unused.
Leave AD0 low/default for address 0x68.
Keep stem and pot touch pads separate.
Check rail continuity; bridge any split rails.
Module pin positions vary: follow printed labels.

All sounds come from the laptop.

References: Arduino UNO R3 + Adafruit MPR121 / MPU6050 pinouts